14 Configuration and secrets management

This chapter covers

- Configuring applications on Kubernetes
- Using ConfigMaps and Secrets in Kubernetes
- Managing deployments and configuration with Kustomize

Releasing applications to production involves two important aspects: an executable artifact and its configuration. The executable artifact could be a JAR file or a container image. The previous chapters covered several principles, patterns, and tools for building applications that are loosely coupled, resilient, scalable, secure, and observable. You saw how to package applications as executable JAR artifacts or container images. I also guided you through the implementation of the commit stage of a deployment pipeline, which ultimately produces a release candidate.

The other aspect of being ready for production is configuration. Chapter 4 introduced the importance of externalized configuration for cloud native applications and covered several techniques for configuring Spring Boot applications. This chapter will continue that discussion in preparation for deploying an entire cloud native system to a Kubernetes production environment.

First I'll describe a few options for configuring Spring Boot applications on Kubernetes and describe what's missing for using Spring Cloud Config in production. Then you'll learn how to use ConfigMaps and Secrets, a native mechanism for handling configuration on Kubernetes. As part of the